Communications and Networks

Baseband Waveform

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Recall the Source Coding

Sampling

Quantization

Compressive Coding

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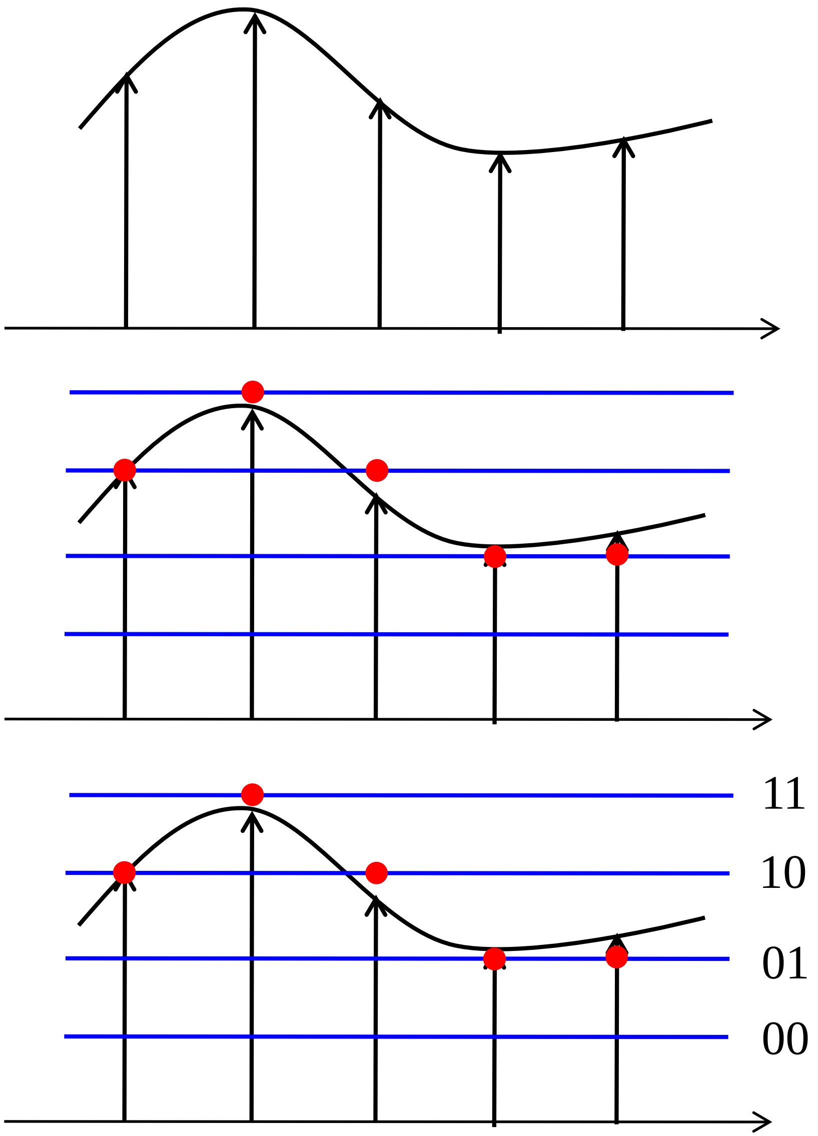
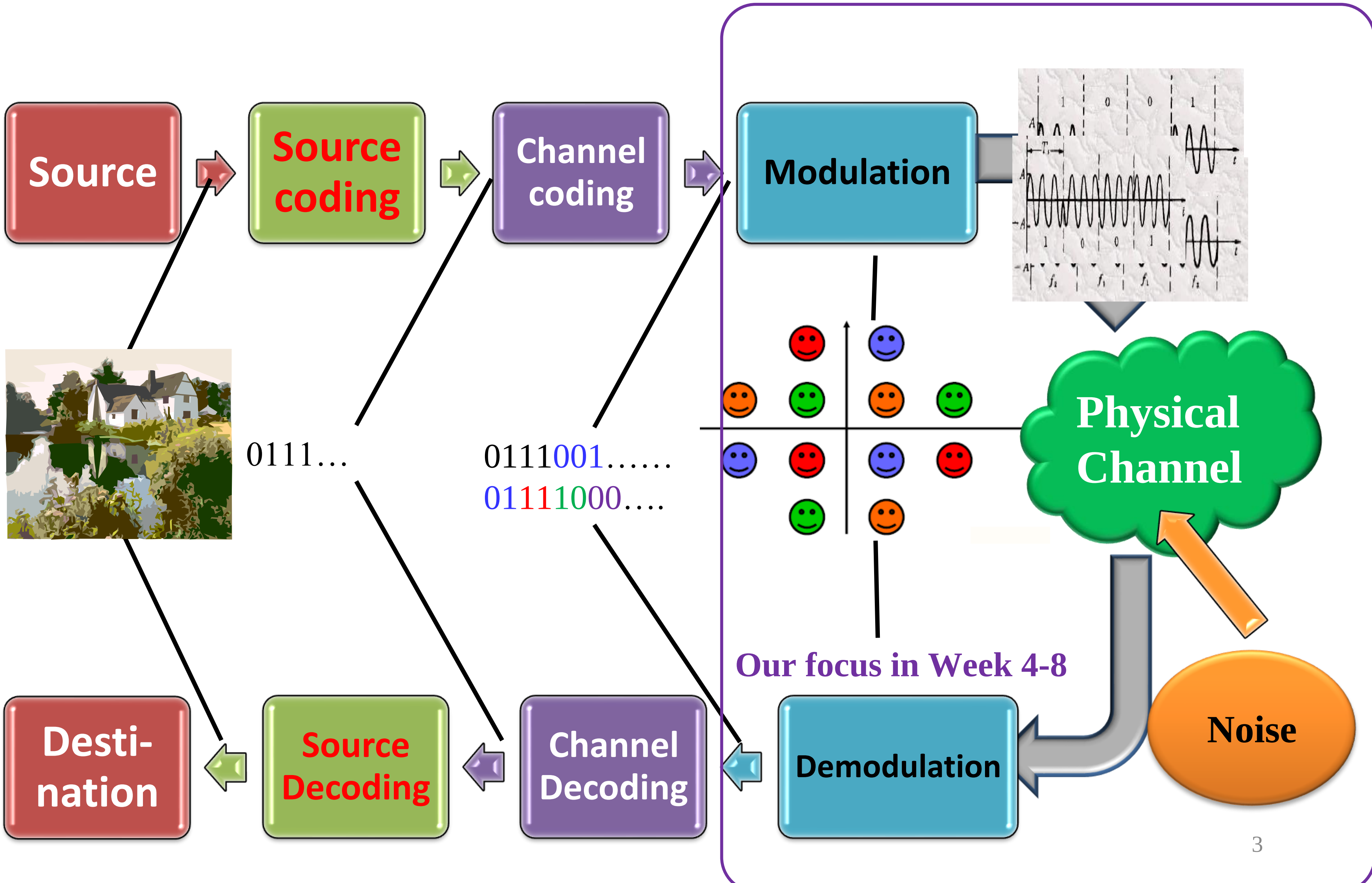


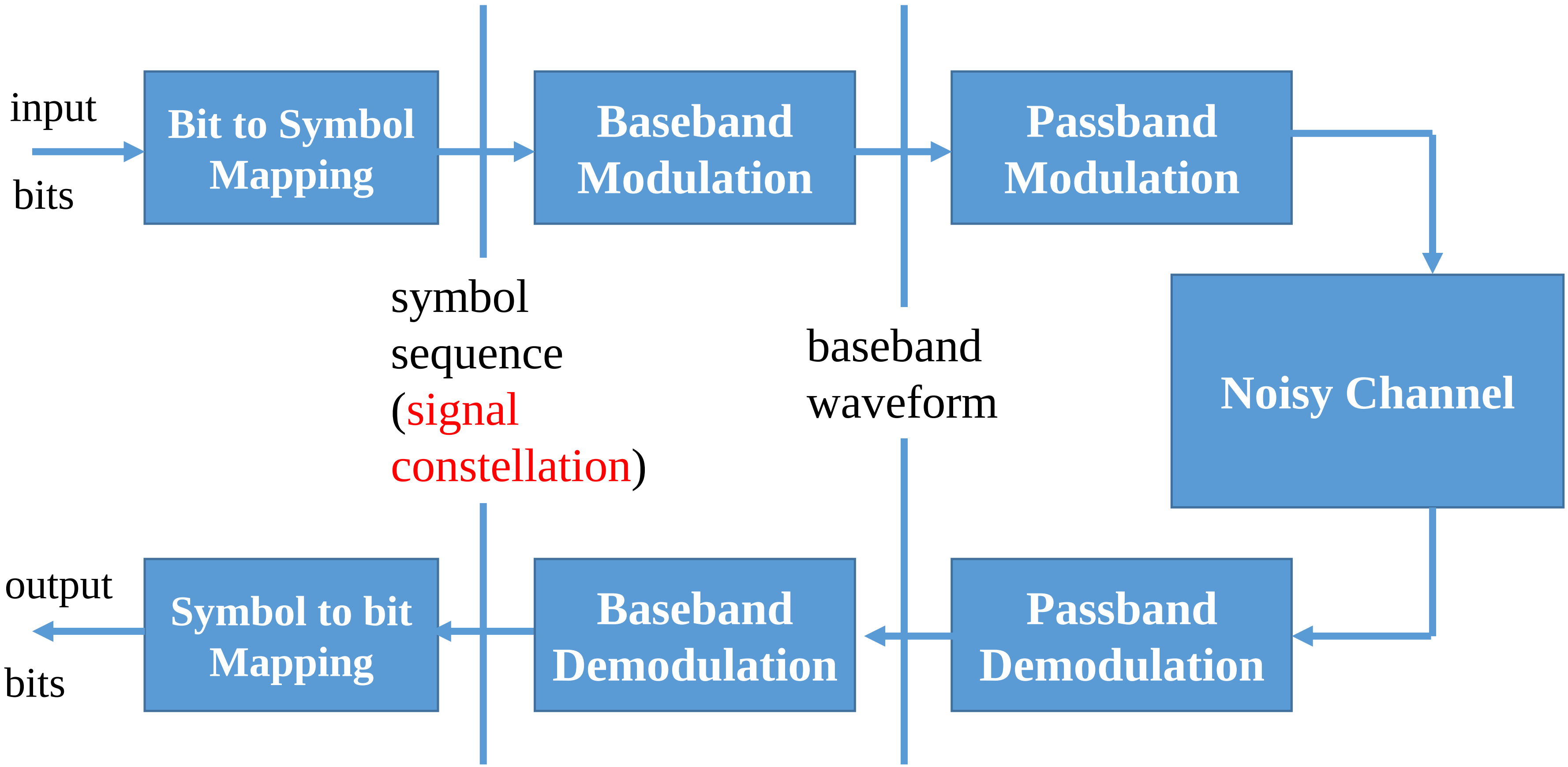
Digital Communication Link: Building Blocks



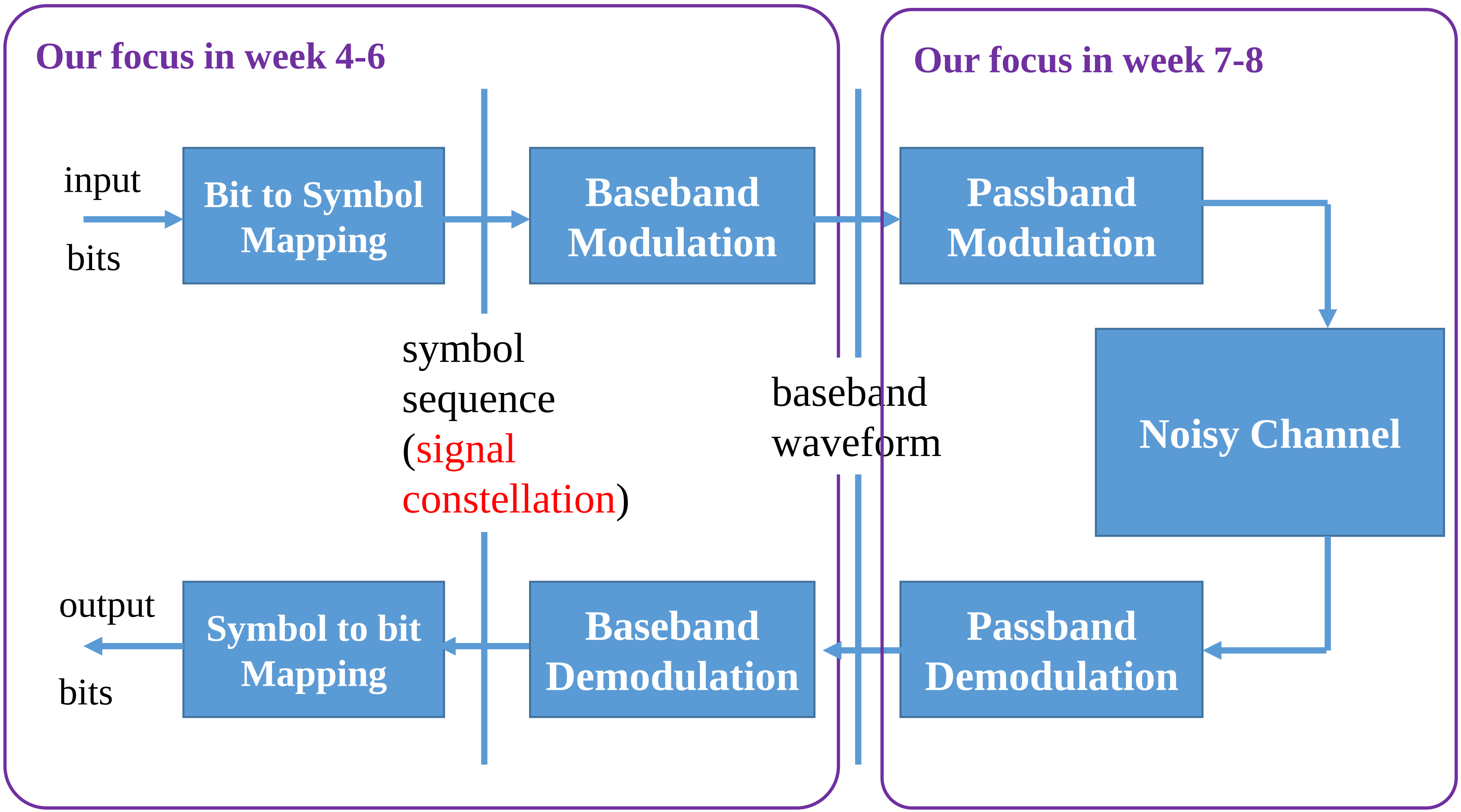
Outline

- Overview of Baseband Transmission
- Bit to Symbol (Constellation) Mapping
- Baseband Waveform
  - Nyquist Criterion
  - Raised Cosine Filter

Modulation: Overview



Modulation: Overview



Modulation: Concept

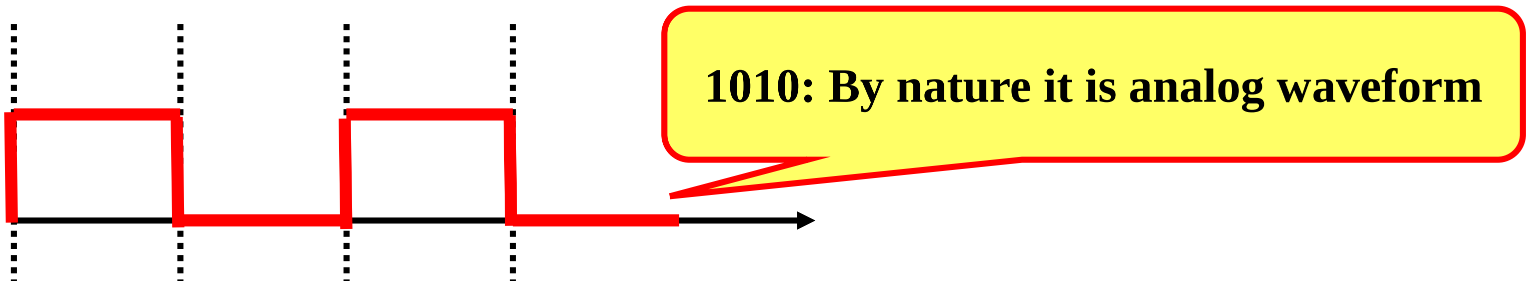
- Modulation:
  - Maps the digital information into analog waveforms that match the characteristics of the channel

- Why analog waveform?
  - The physical world is analog

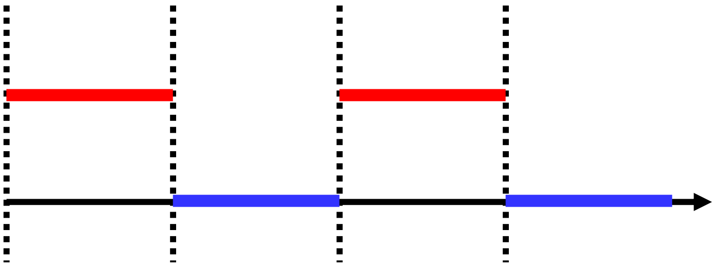
  - The channel has bandwidth limitations

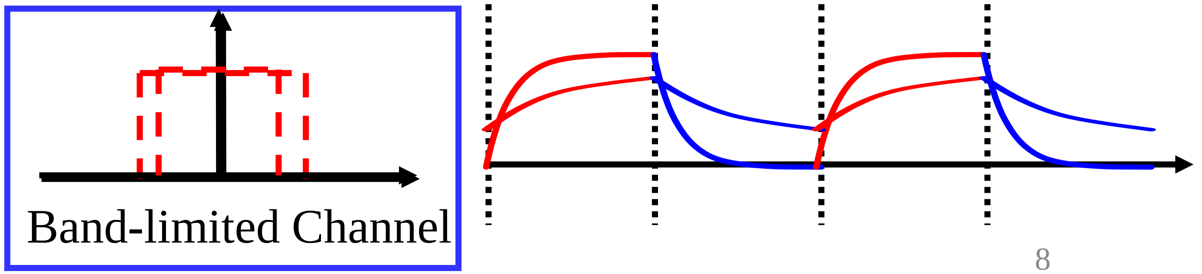
We Can Only Transmit Analog Signals Over Limited Bandwidth

- You cannot "directly" transmit bits over the physical world
  - Even if the simplest way you can think of:



- Also the channel is band-limited
  - The waveform design should match the bandwidth of the channel, otherwise distortion occurs





How to Design the Waveform?

- Recall sampling: $x(t) \rightarrow x(nT_s)$

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \frac{\sin 2\pi f_H (t - nT_s)}{2\pi f_H (t - nT_s)}$$

- Now *reversely*: bits → symbol sequence a_k → $s(t)$

$$\boxed{s(t) = \sum_k a_k \phi_k(t)}$$

symbol modulated on waveform

How to provide the symbol sequence a_k from the bit sequence?

How to design the waveform function?

Baseband Transmission

- Baseband: the waveform has low-pass spectrum
  - Suitable for the low-pass channel

- We will cover:
  - Bit to symbol mapping $\boxed{a_k}$
  - Waveform design

$$\boxed{\phi_k(t)}$$

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Bit Sequence to Symbols

… 01100011111010111110110010101100…

a_1 a_2 a_3 a_4 a_5 a_6 a_7 a_8

- Packing r bits into one symbol
- Symbol a_k is in a finite set

$$a_k \in \mathcal{A}$$

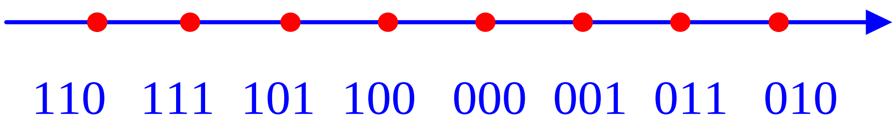
$$M =: |\mathcal{A}| = 2^r \quad \Rightarrow \quad r = \log_2 |\mathcal{A}|$$

M: number of possible symbols in the symbol set (alphabet)

Forms of Transmission Symbols (1)

- They can be real numbers

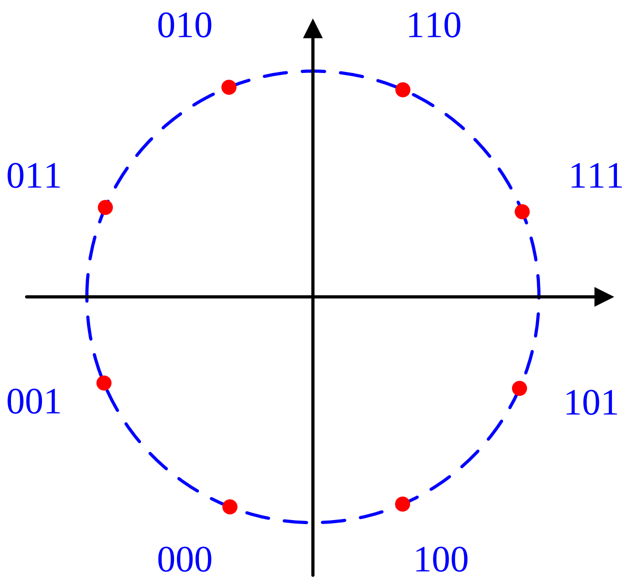
Amplitude **Shift Keying (ASK)**



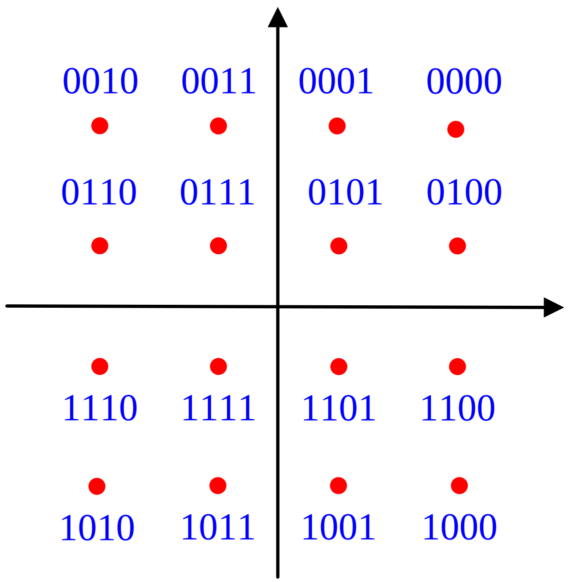
Pulse *Amplitude* Modulation（PAM）
with M = 8, r = 3 bits

Forms of Transmission Symbols (2)

- They can be complex numbers
  - Constellation



Phase Shift Keying (PSK)



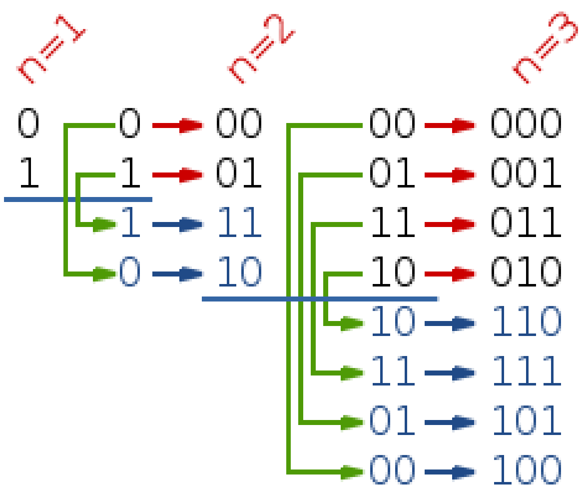
Quadrature Amplitude Modulation (QAM)

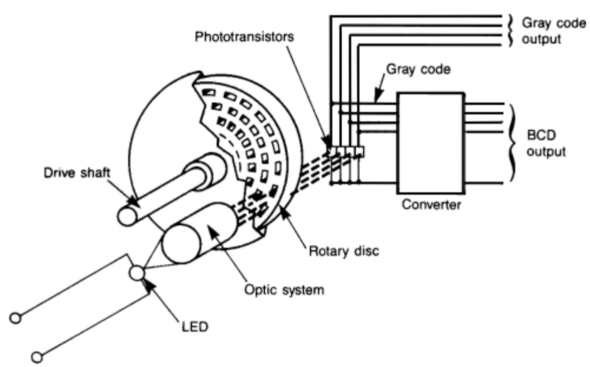
Gray Encoding

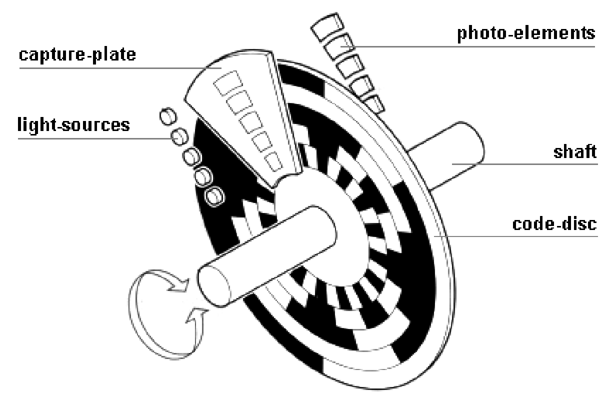
- The mapping of the r-bits to the M possible symbols, i.e.,

$$\boxed{\mathcal{G} : \{0,1\}^{\log_2|\mathcal{A}|} \mapsto \mathcal{A}}$$

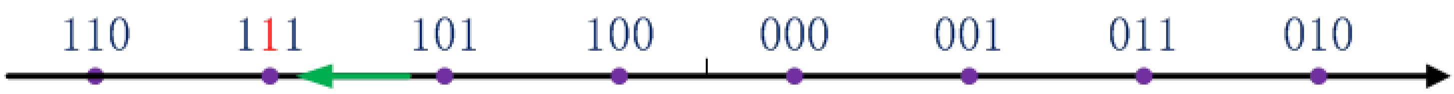
  can be done in a number of ways, in fact arbitrary
- One good choice: Gray encoding:
  - **Adjacent symbols only differ by ONE bit** (WHY good?)



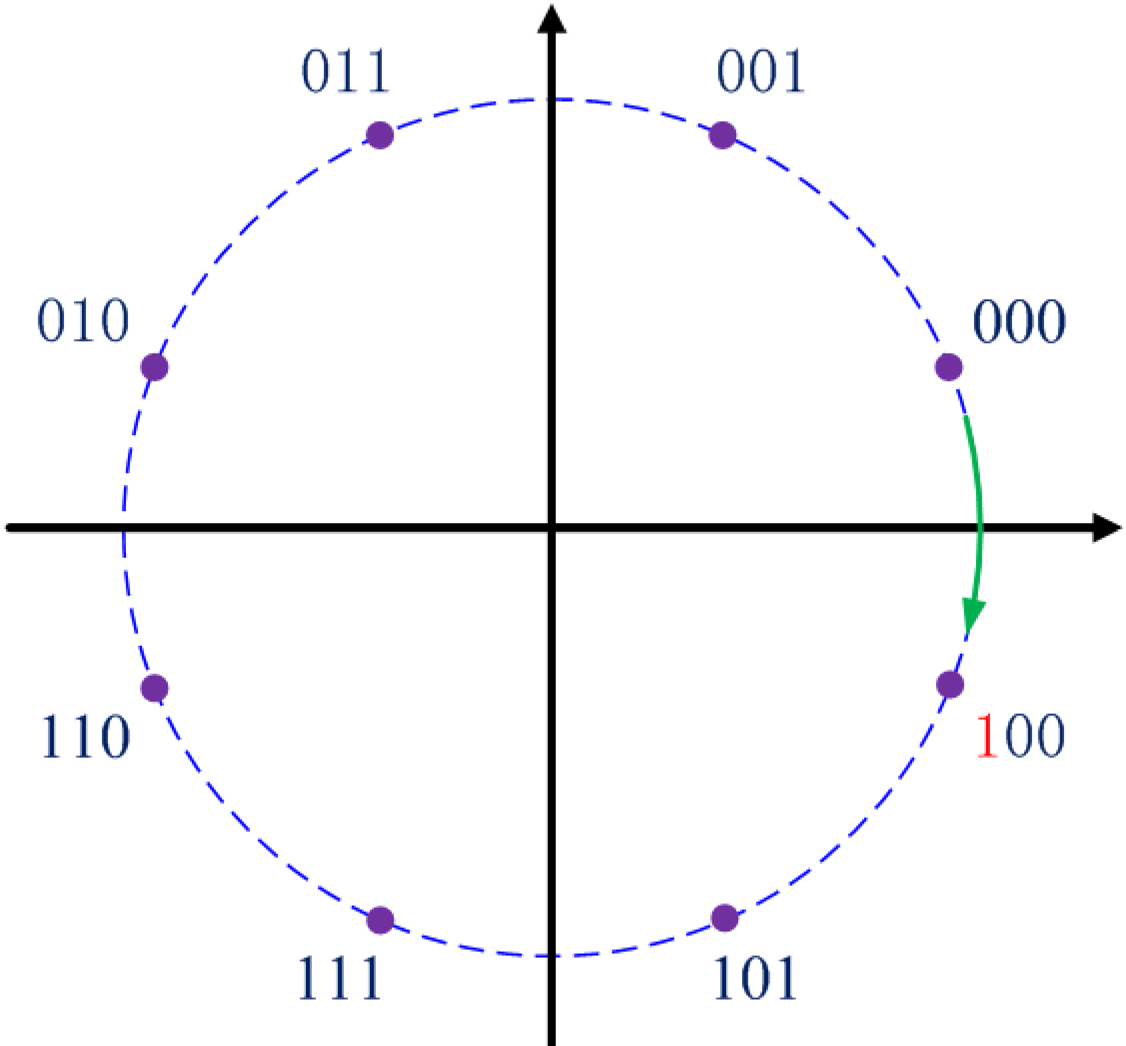




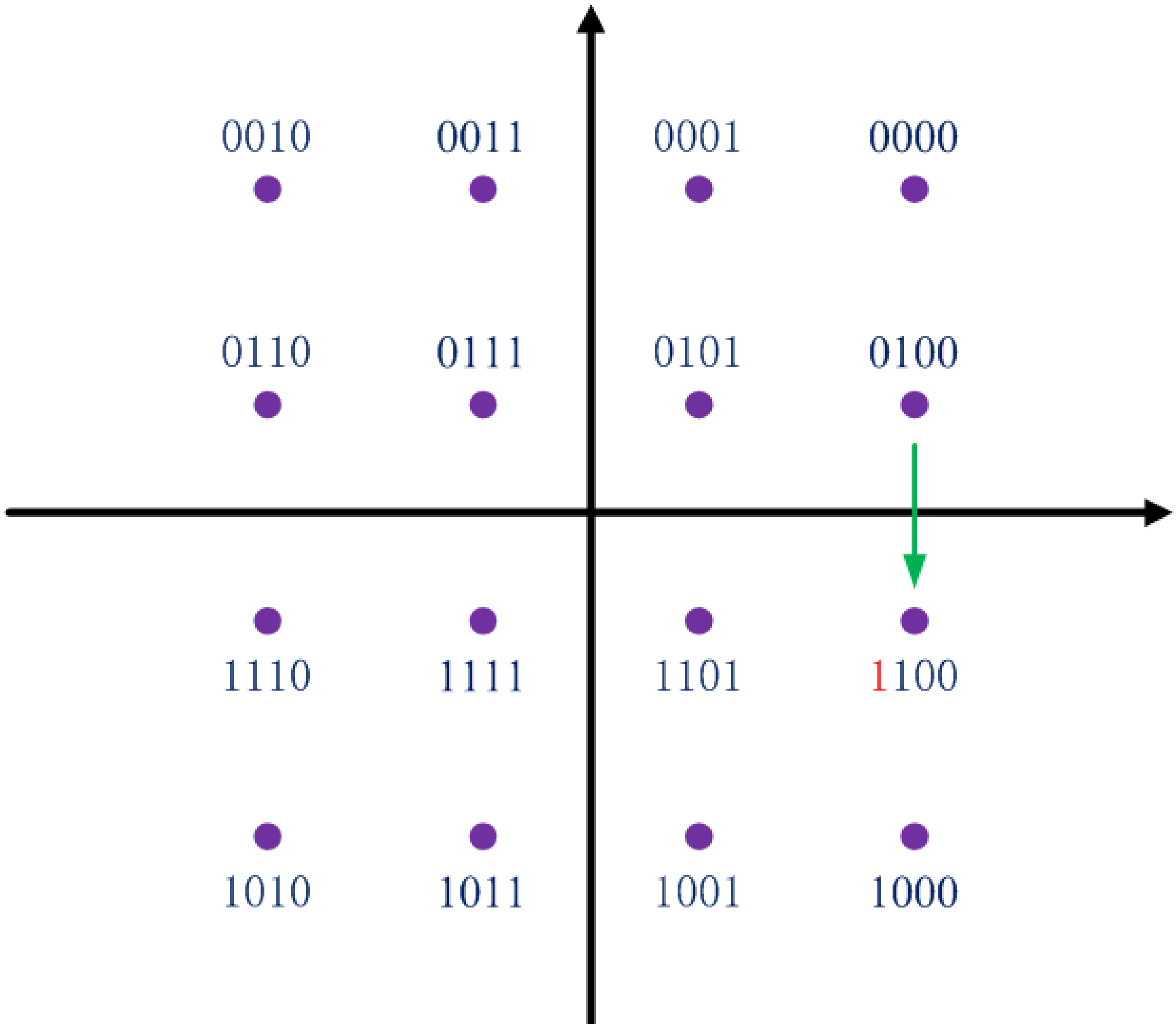
Gray Encoding for PAM



Gray Encoding for PSK



Gray Encoding for QAM



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Baseband Modulation: Overview

- Use the same pulse waveform for all symbols
  - With time shifts, i.e., send a symbol every T_s seconds

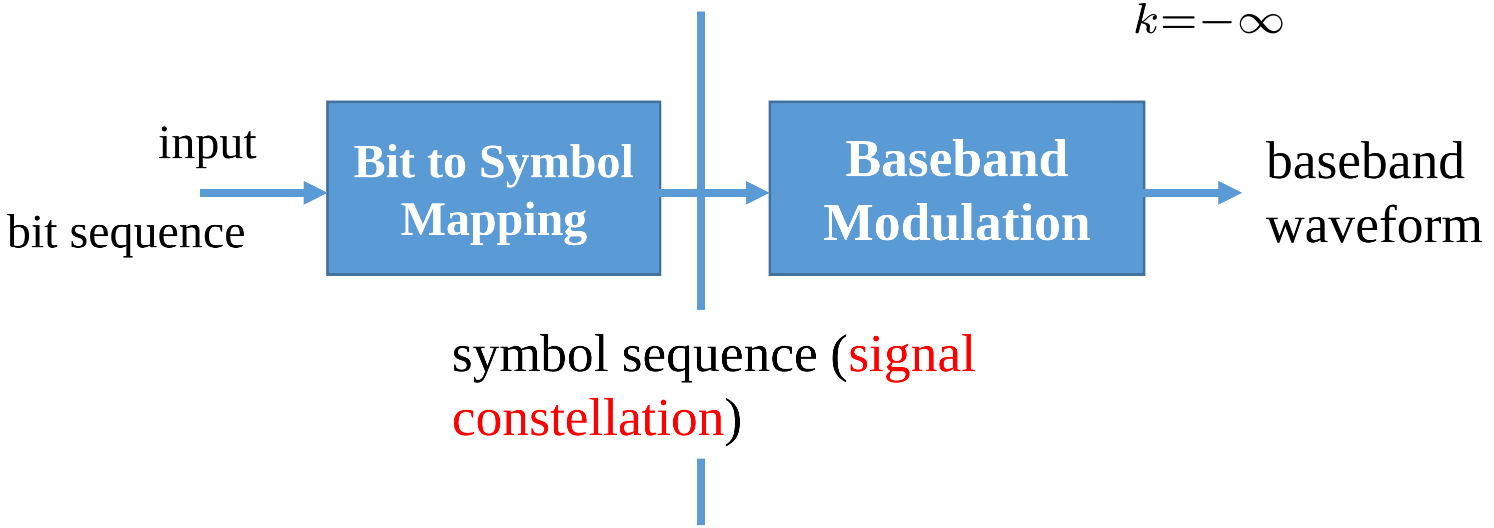
$$\phi_k(t) = g(t - kT_s)$$

- Our goal: decide T_s and $g(t)$

0110001…

$a_1, a_4, a_3, \ldots$

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$



Concept of Symbol Rate

- Send a symbol every T_s seconds
- Symbol rate, number of symbols transmitted per second

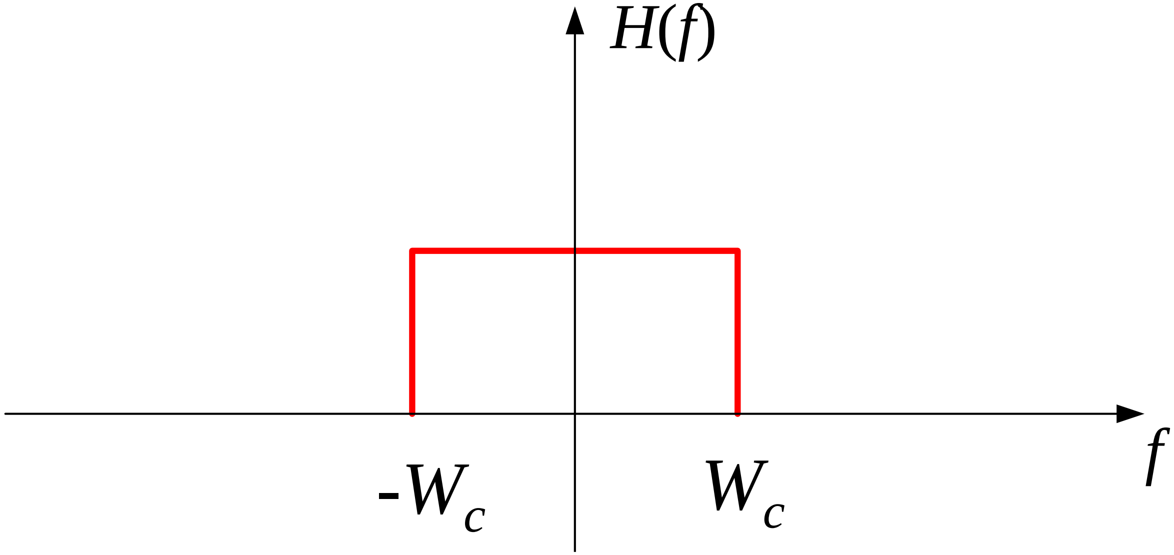
$$\boxed{R_s = \frac{1}{T_s}}$$

- Bit rate, number of bits transmitted per second

$$\boxed{R_b = R_s \log_2 M = \frac{1}{T_s} \log_2 M}$$

Requirement of g(t)

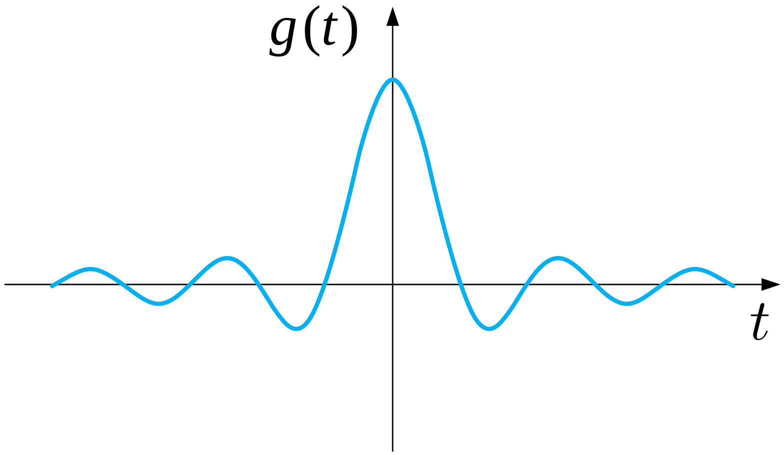
- Characteristic of the channel: Band-limited!
  - Spectrum of the (low-pass) channel (impulse response):

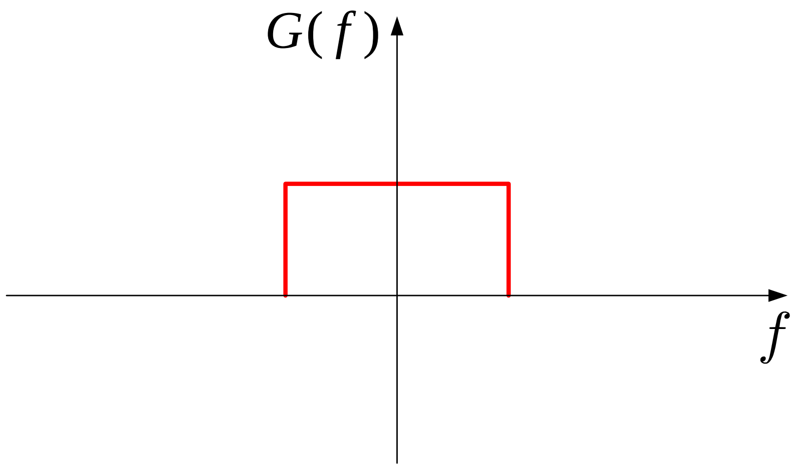


The bandwidth of the channel is W_c

Requirement of g(t): Bandlimited

- For example:





$$\boxed{g(t) = \frac{\sin 2\pi W t}{2\pi W t}} \quad \overset{\mathcal{F}}{\Longleftrightarrow} \quad \boxed{G(f) = \begin{cases} 1/2W & |f| \le W \\ 0 & |f| > W \end{cases}}$$

As long as *W<=W_c*, no distortion as the waveform signal passes through the channel low-pass channel

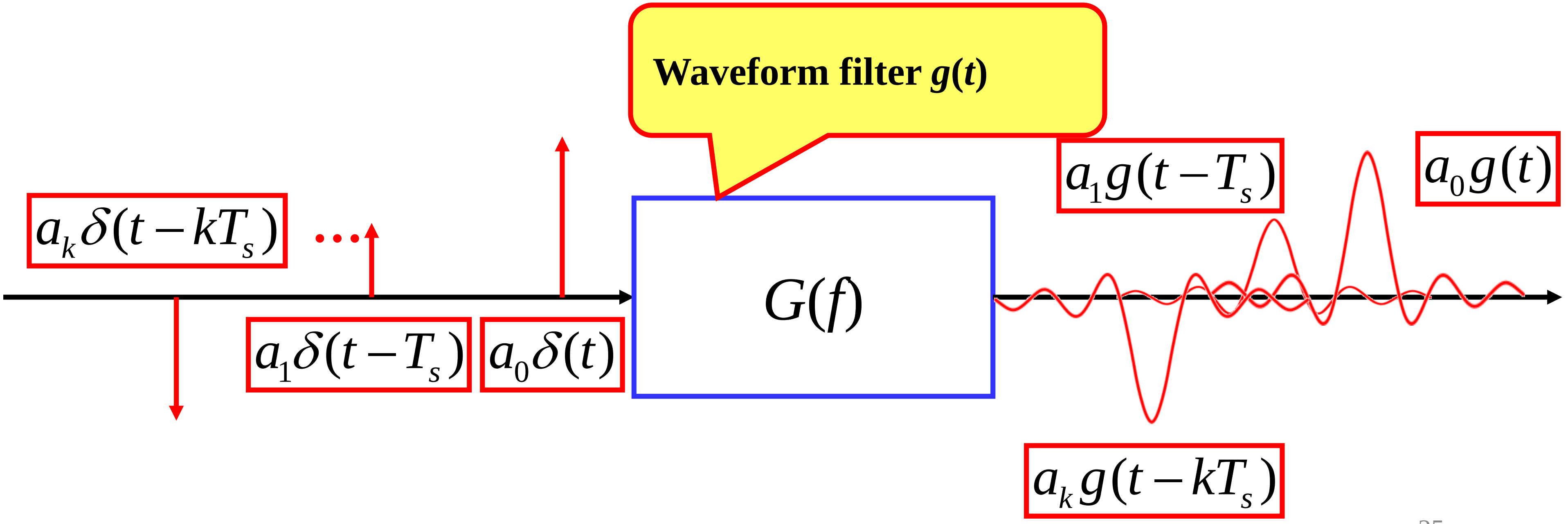
Putting Together

- If $g(t)$ is band-limited
- Is $s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$ band-limited?

Generation of s(t)

$$\boxed{s(t)=\sum_{k=-\infty}^{\infty} a_k g(t-kT_s)}$$

- Let the pulses $\boxed{a_k\delta(t-kT_s)}$ pass through a low-pass filter whose impulse response is $g(t)$



Not done yet…

- Two key questions remain:

  - Is $g(t) = \dfrac{\sin 2\pi W t}{2\pi W t}$ the only choice?

  - Any connection between W and T_s ?
    - Alternatively, given the baseband waveform bandwidth W (being less than the channel bandwidth), how to set T_s ?

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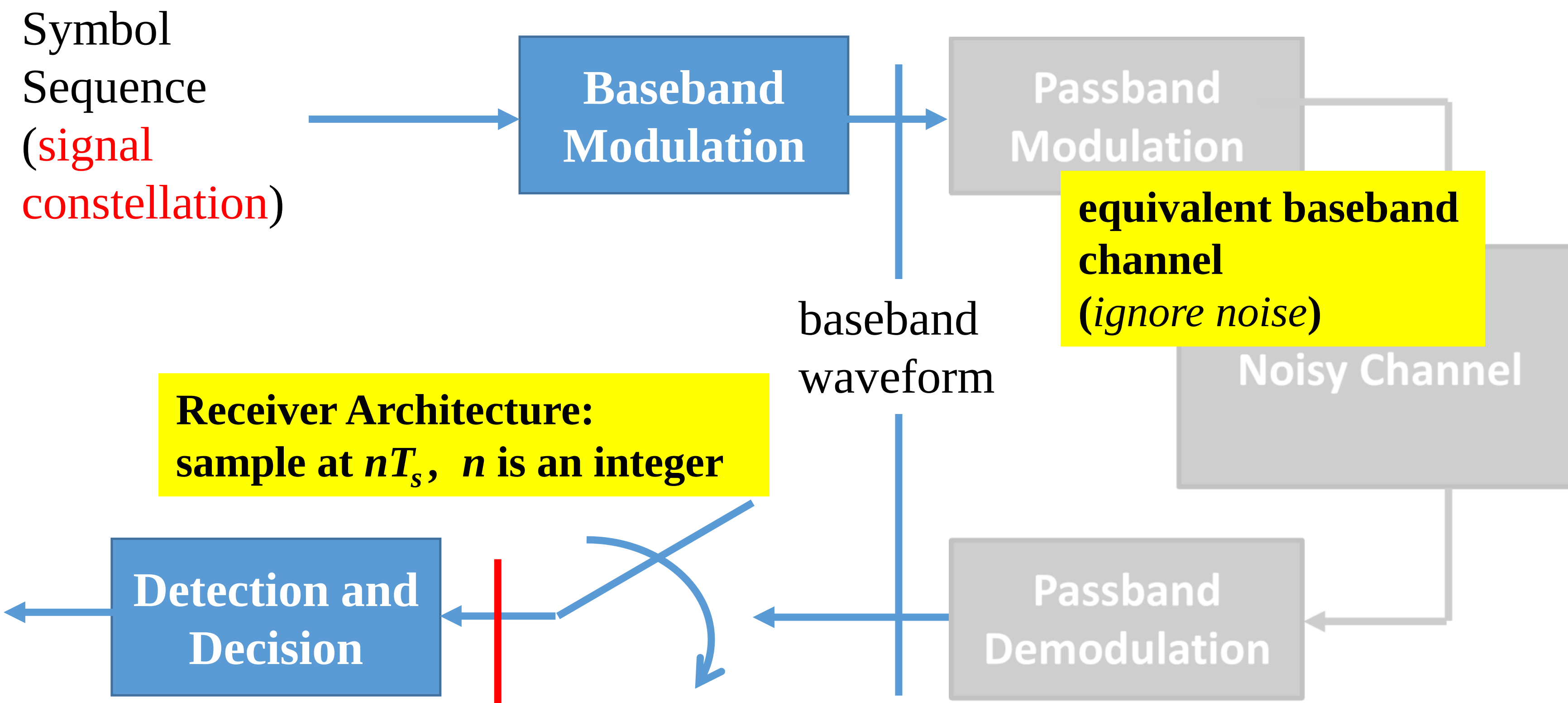
Receiver Architecture

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$



$$s(nT_s) = \sum_{k=-\infty}^{\infty} a_k g(nT_s - kT_s)$$

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$

Inter-symbol Inference

- Receiver samples at times nT_s

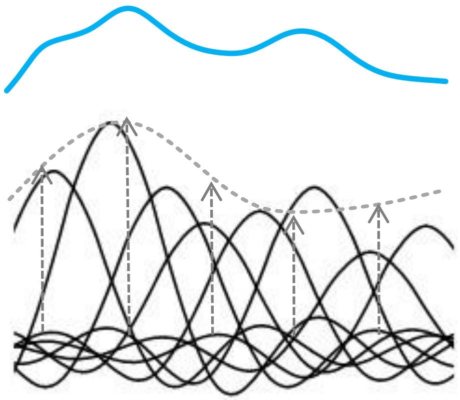
$$\boxed{s(t)=\sum_{k=-\infty}^{\infty} a_k g(t-kT_s)}$$

Sample at nT_s

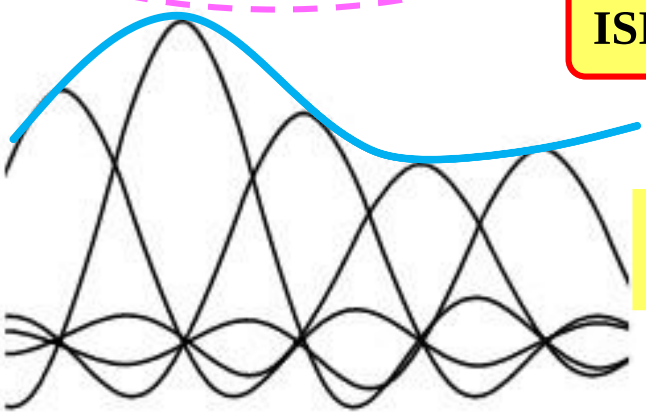
$$\boxed{s(nT_s)=a_n g(0)+\sum_{k=-\infty,k\neq n}^{\infty} a_k g\big((n-k)T_s\big)}$$

Inter-Symbol Interference (ISI)

ISI should be 0



$$\boxed{s(nT_s)=a_n}$$



Condition for 0 ISI?

Conditions for no ISI (1)

- Two conditions for $g(t)$

$$\boxed{g(0)=1}$$

$$\boxed{\sum_{k=-\infty,k\neq n}^{\infty} a_k g\left((n-k)T_s\right)=0} \Leftrightarrow \boxed{\sum_{k'=-\infty,k'\neq 0}^{\infty} a_{n-k'} g(k'T_s)=0}$$

- Should hold for any values of a_k, therefore

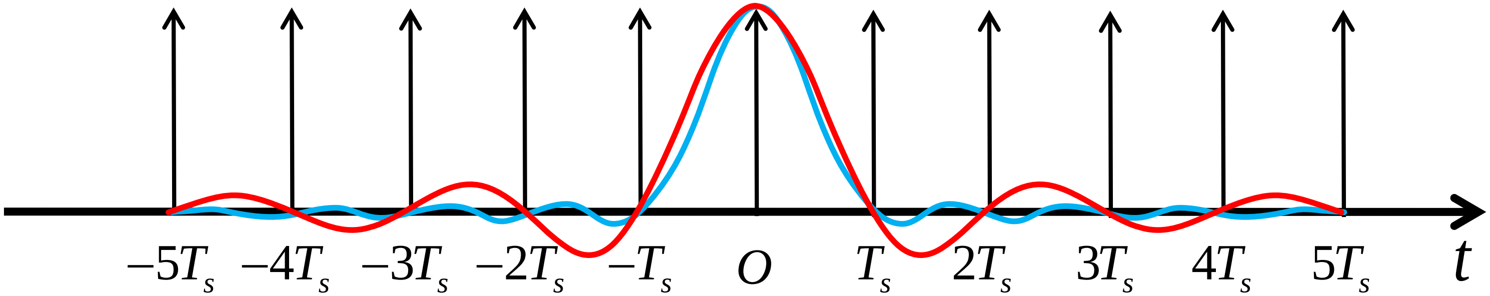
$$\boxed{g(kT_s)=\begin{cases}1 & k=0\\ 0 & k\neq 0\end{cases}}$$

Conditions for no ISI (2)

- There can be multiple choices

$$\boxed{g(kT_s) = \begin{cases} 1 & k = 0 \\ 0 & k \neq 0 \end{cases}}$$

Only those times nT_s matter



- which is equivalent to

$$\boxed{g(t)\sum_{n=-\infty}^{\infty}\delta(t-nT_s) = \delta(t)}$$

Nyquist Criterion

- Look at the spectrum

$$\boxed{g(t)\sum_{n=-\infty}^{\infty}\delta(t-nT_s)=\delta(t)}$$

$$\Updownarrow \mathcal{F}$$

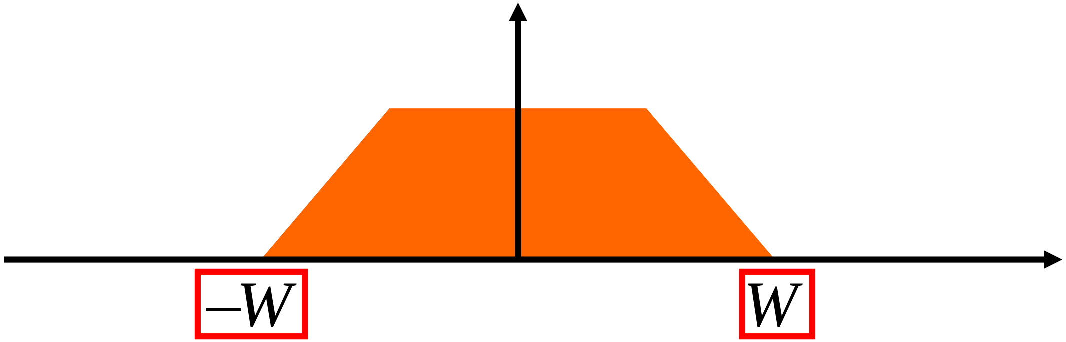
$$\boxed{G(f)*\sum_{n=-\infty}^{\infty}\frac{1}{T_s}\delta\left(f-\frac{n}{T_s}\right)=1}$$

$$\boxed{\sum_{n=-\infty}^{\infty}G\left(f-\frac{n}{T_s}\right)=T_s}$$

Shift the replicas of the spectrum of *g*(*t*) by n/T_s , where $n \in \mathbb{Z}$
The resultant spectrum is a constant, if and only if no-ISI

More about Nyquist Criterion (1)

- Suppose the bandwidth of $g(t)$ is W

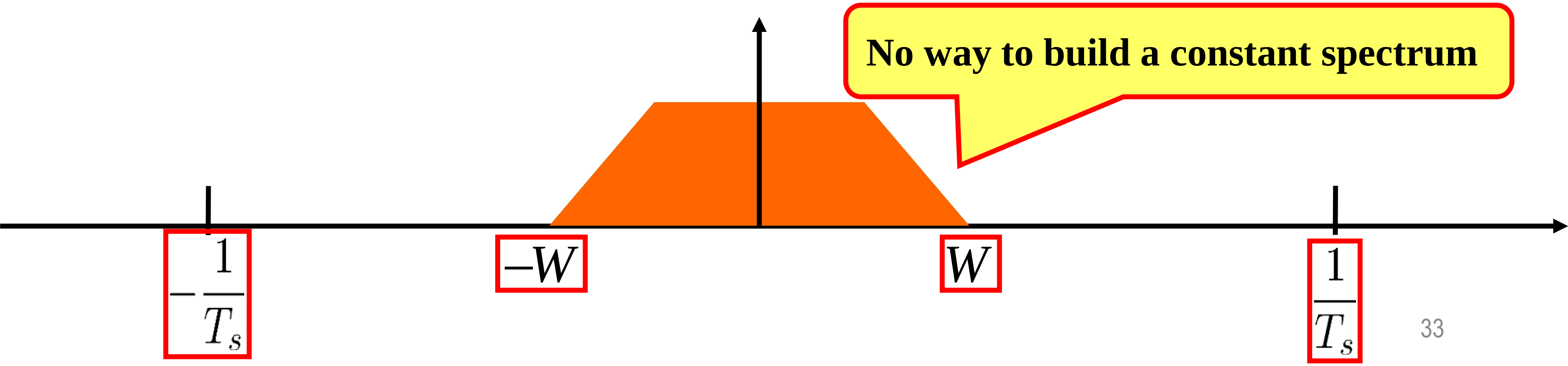


- Case 1:

$$\boxed{T_s < \frac{1}{2W} \quad \Leftrightarrow \quad \frac{1}{T_s} > 2W}$$

What happens in the time domain?

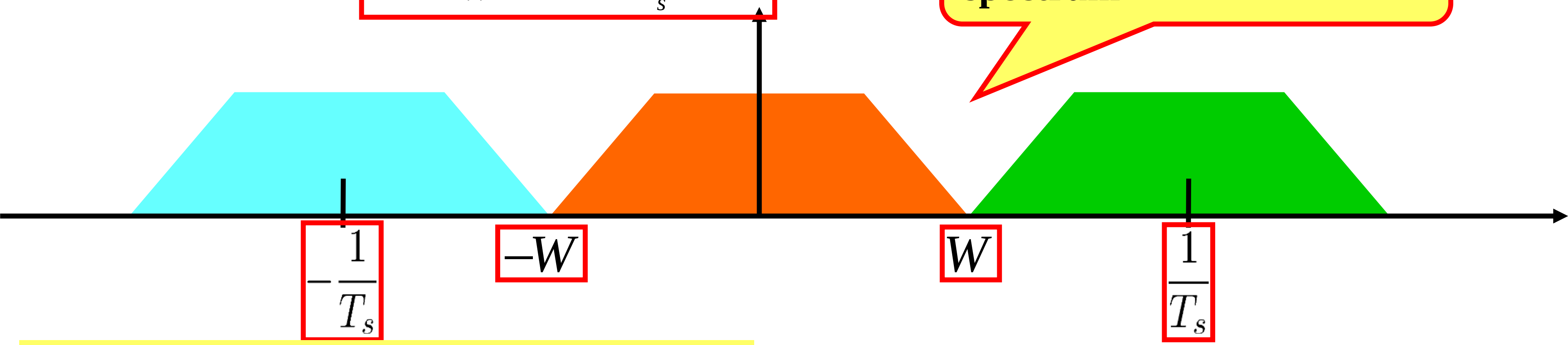
No way to build a constant spectrum



More about Nyquist Criterion (2)

- Case 2: $T_s = \frac{1}{2W} \Leftrightarrow \frac{1}{T_s} = 2W$

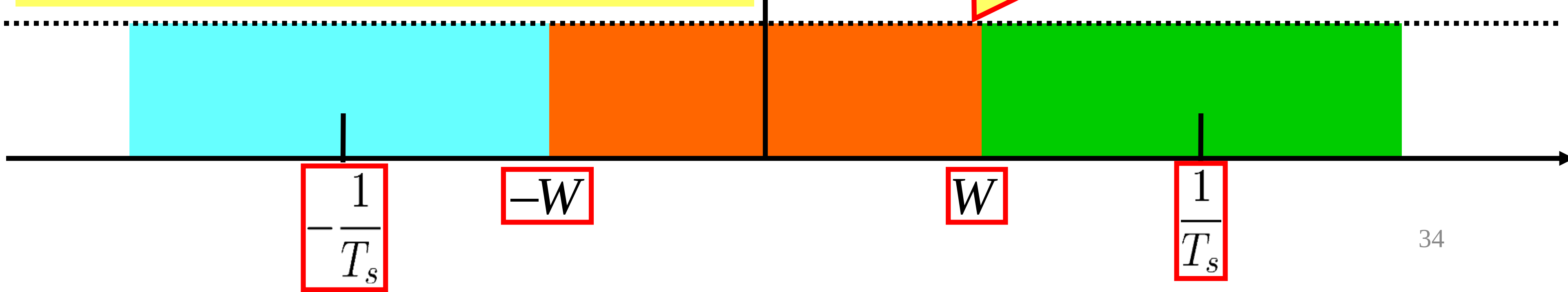
Still cannot build constant spectrum



The minimum T_s is $1/(2W)$

The maximum symbol rate R_s is $2W$

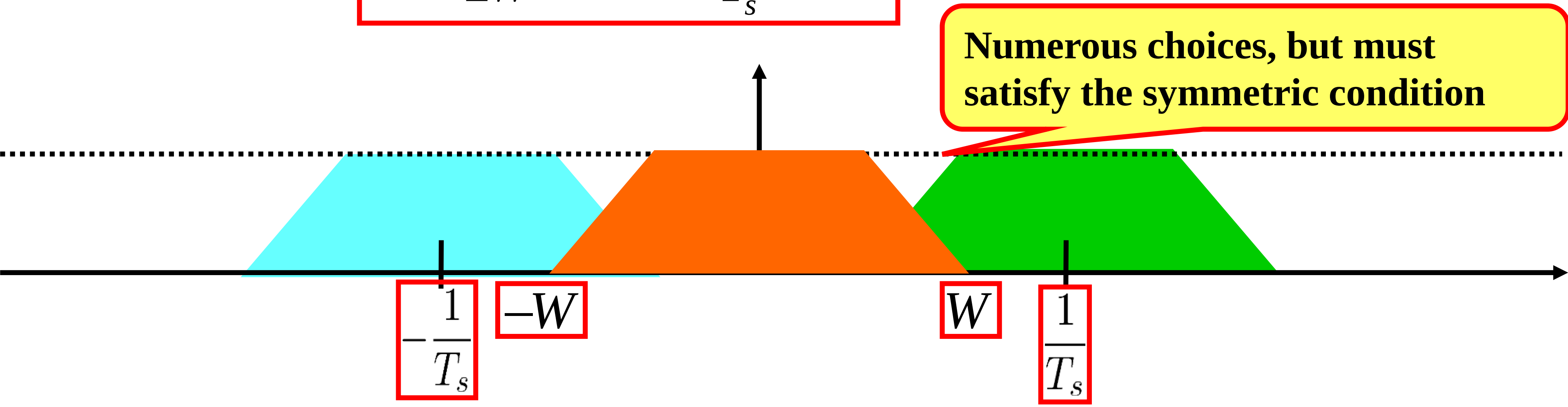
In this case, the only choice of g(t) is the ideal low-pass filter (sinc function)



More about Nyquist Criterion (3)

- Case 3：

$$\boxed{T_s > \frac{1}{2W} \quad \Leftrightarrow \quad \frac{1}{T_s} < 2W}$$



- In **conclusion**
  - The symbol rate is limited by the baseband bandwidth

$$\boxed{R_s = \frac{1}{T_s} \le 2W}$$

Data Rate & Bandwidth Efficiency

- The symbol rate must satisfy

$$\boxed{R_s \le 2W}$$

- We also have bit $\boxed{R_b = R_s \log_2 M}$

- Therefore $\boxed{R_b \le 2W \log_2 M}$ **"Broadband Access"!**

- Bandwidth efficiency: data rate per bandwidth

$$\boxed{\eta = \frac{R_b}{W} \le 2\log_2 M}$$

"Broadband Access"!

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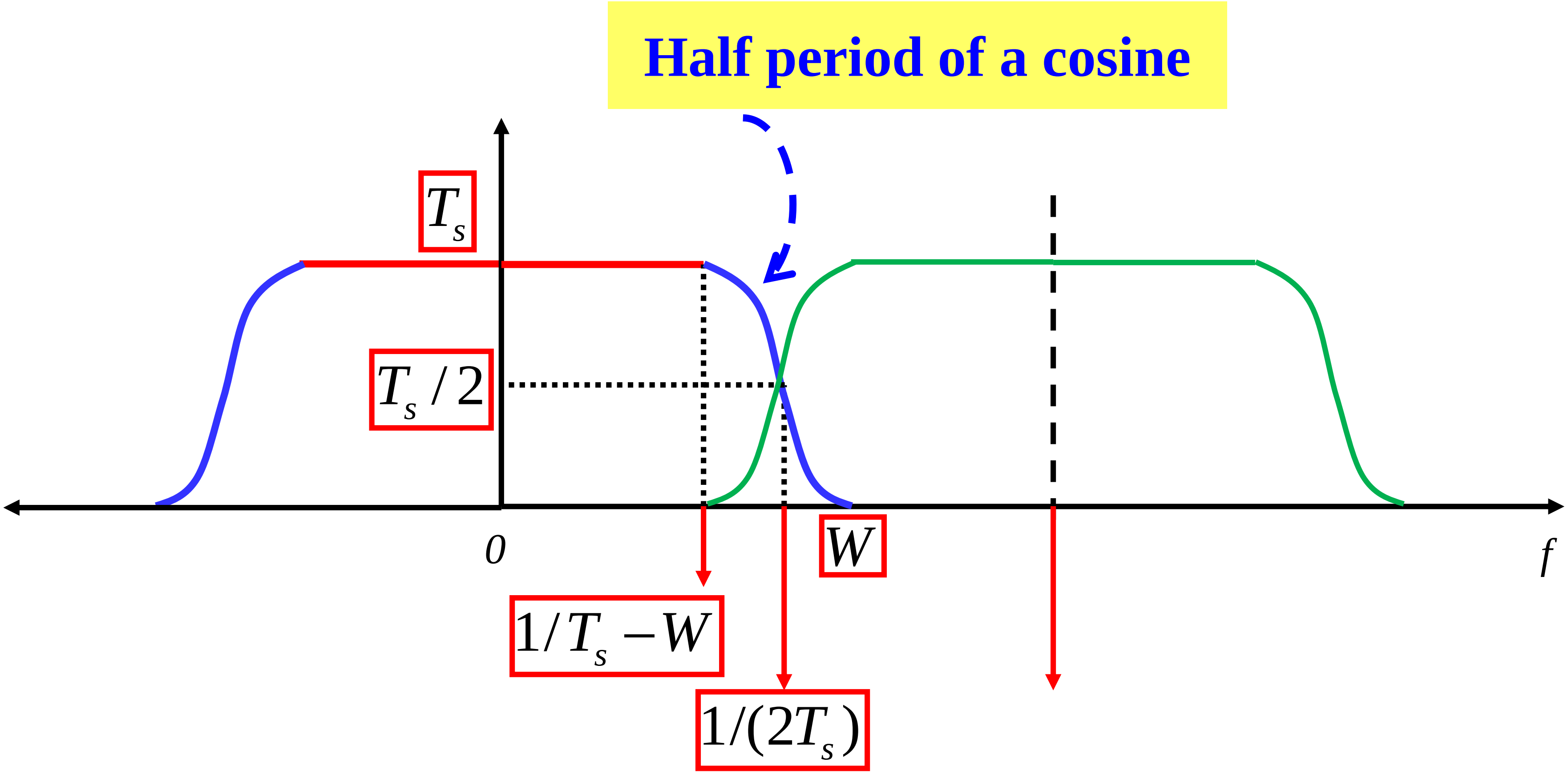
Raised Cosine Filter (1)

- Ideal low-pass filter is hard to realize
- One practical file that can adjust from W to 2W
  - Raised Cosine（Used in GSM systems), its spectrum G(f)

$$H(f)=\begin{cases} T_s & 0\le |f| < \dfrac{1-\alpha}{2T_s} \\ \dfrac{T_s}{2}\left\{1+\cos\left[\dfrac{\pi T_s}{\alpha}\left(|f|-\dfrac{1-\alpha}{2T_s}\right)\right]\right\} & \dfrac{1-\alpha}{2T_s}\le |f| \le \dfrac{1+\alpha}{2T_s} \\ 0 & |f| > \dfrac{1+\alpha}{2T_s} \end{cases}$$

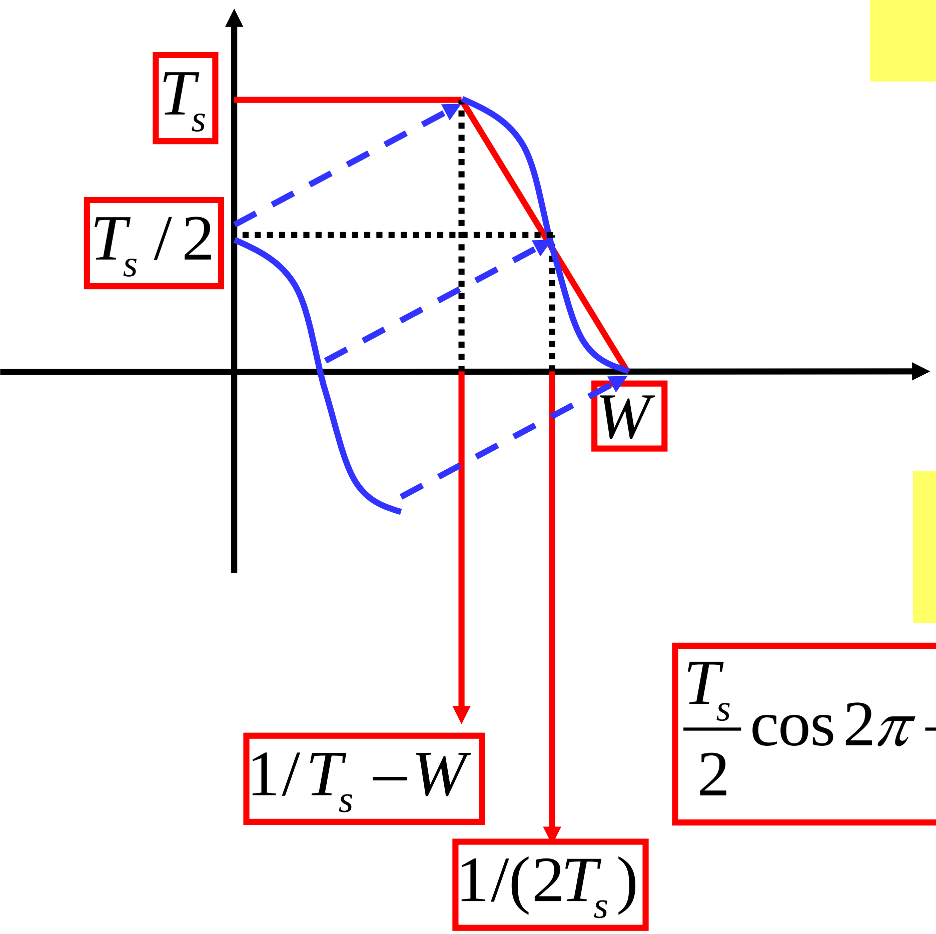
Raised Cosine Filter (2)

- Details of the filter spectrum



Raised Cosine Filter: Expression (1)

- Expression of the cosine part:



First decide period of cosine

$$\frac{\tau}{2} = W - (1/T_s - W) = 2W - 1/T_s$$

$$\tau = 4W - 2/T_s$$

Therefore the expression of the *original* cosine function:

$$\frac{T_s}{2}\cos 2\pi\frac{f}{\tau} = \frac{T_s}{2}\cos\frac{\pi f}{2W - 1/T_s} = \frac{T_s}{2}\cos\frac{\pi T_s f}{2WT_s - 1}$$

Raised Cosine Filter: Expression (2)

- The original cosine function:

$$\frac{T_s}{2}\cos\frac{\pi T_s f}{2WT_s - 1}$$

- Define:

$$\alpha = 2WT_s - 1$$

rolloff factor

$$\frac{T_s}{2}\cos\frac{\pi T_s f}{\alpha}$$

- Raise the cos
  - Upward by $T_s/2$

$$\frac{T_s}{2}+\frac{T_s}{2}\cos\frac{\pi T_s f}{\alpha}$$

  - Rightward by $1/T_s$-W

$$\frac{T_s}{2}+\frac{T_s}{2}\cos\frac{\pi T_s\left[f-(1/T_s-W)\right]}{\alpha}$$

Raised Cosine Filter: Expression (3)

- Utilizing the relation:

$$\boxed{\frac{1}{T_s} - W = \frac{1-\alpha}{2T_s}}$$

We now have the rolloff part of the filter for the positive frequencies

$$\boxed{\begin{aligned}\frac{T_s}{2} + \frac{T_s}{2}\cos\frac{\pi T_s\left[f-(1/T_s - W)\right]}{\alpha} &= \frac{T_s}{2} + \frac{T_s}{2}\cos\frac{\pi T_s\left(f-\dfrac{1-\alpha}{2T_s}\right)}{\alpha} \\ &= \frac{T_s}{2}\left\{1+\cos\left[\frac{\pi T_s}{\alpha}\left(f-\frac{1-\alpha}{2T_s}\right)\right]\right\}\end{aligned}}$$

What about the negative frequencies?

Raised Cosine Filter: Expression (4)

- Negative Frequency
  - Upward by $T_s/2$
  - Leftward by $1/T_s$-W

$$\boxed{\frac{T_s}{2}\cos\frac{\pi T_s f}{\alpha}}$$

$$\boxed{\frac{T_s}{2}\left\{1+\cos\left[\frac{\pi T_s}{\alpha}\left(f+\frac{1-\alpha}{2T_s}\right)\right]\right\}}$$

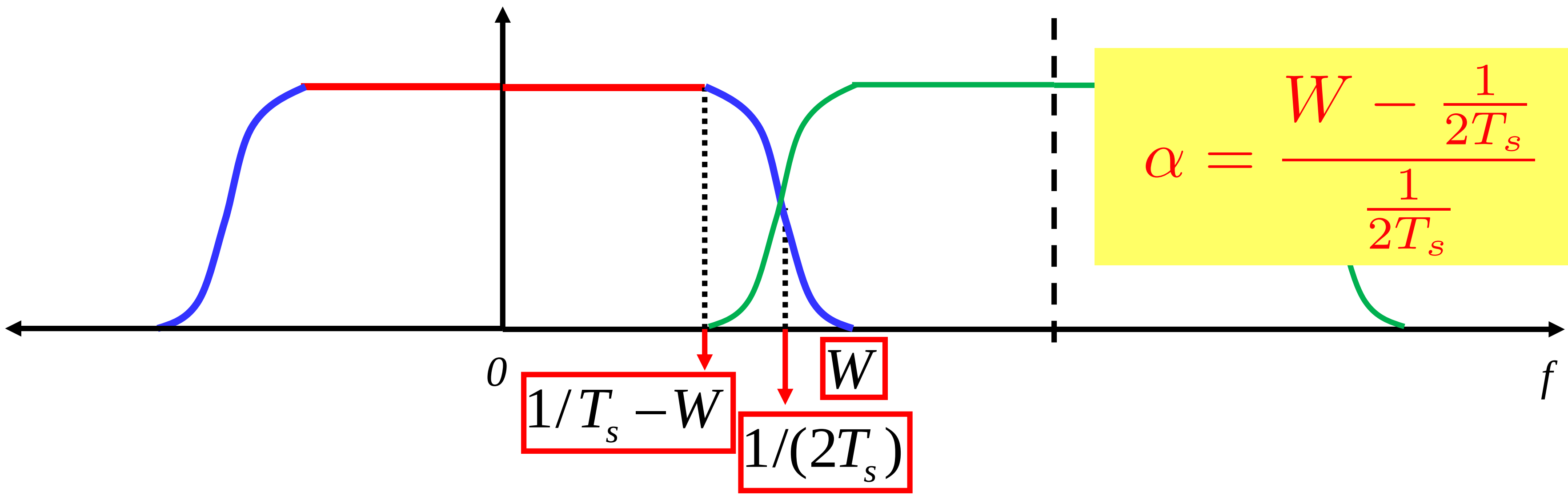
cosine:
even function

$$\boxed{\frac{T_s}{2}\left\{1+\cos\left[\frac{\pi T_s}{\alpha}\left(-f-\frac{1-\alpha}{2T_s}\right)\right]\right\}}$$

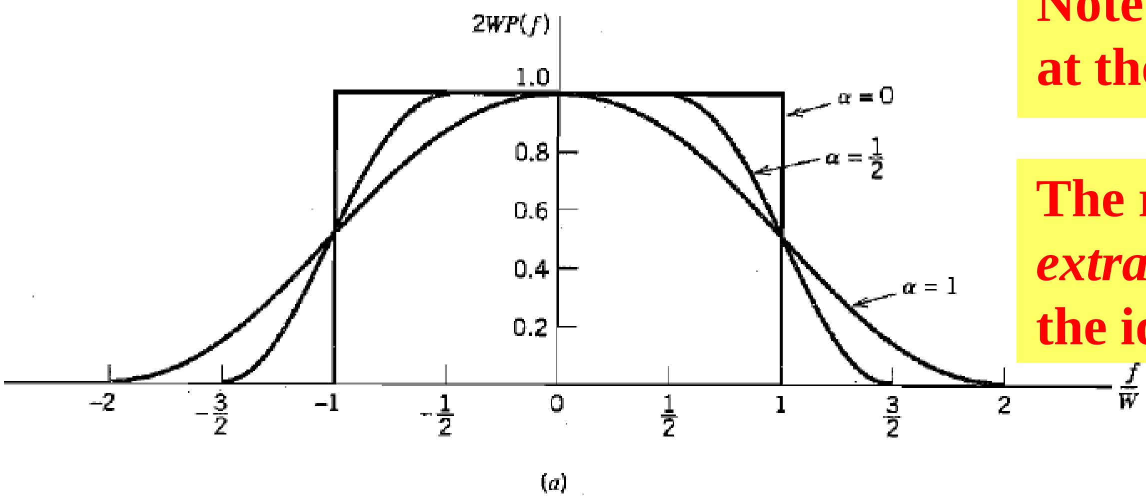
f is negative

$$\boxed{\frac{T_s}{2}\left\{1+\cos\left[\frac{\pi T_s}{\alpha}\left(|f|-\frac{1-\alpha}{2T_s}\right)\right]\right\}}$$

Raised Cosine Filter: Rolloff Factor (1)



$$\alpha = \frac{W - \frac{1}{2T_s}}{\frac{1}{2T_s}}$$



Note that the ideal low-pass filter at the Nyquist rate is $1/(2T_s)$

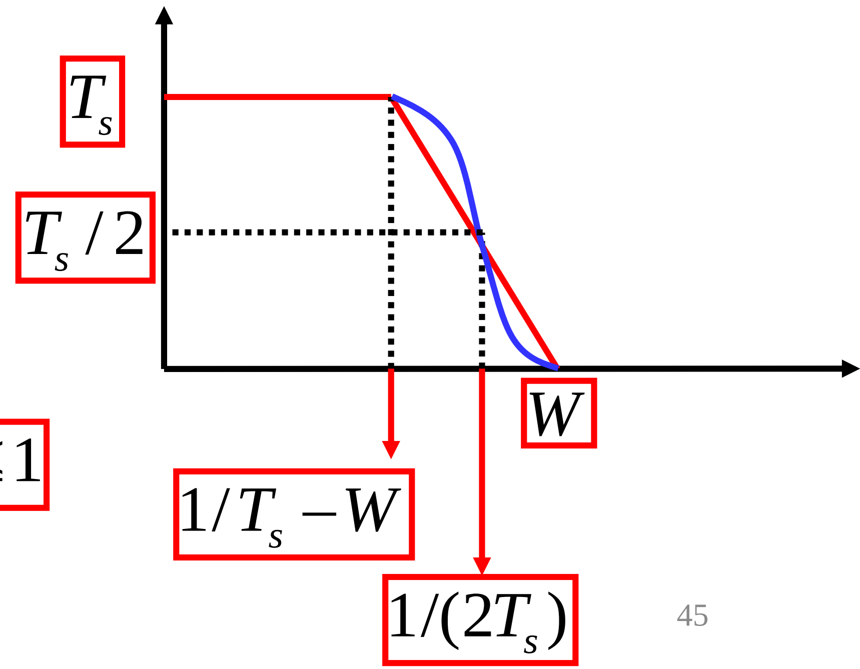
The rolling factor is the unified *extra* bandwidth as compared to the ideal low-pass filter

Raised Cosine Filter: Rolloff Factor (2)

- Feasible values of rolloff factor

$$\boxed{\alpha \in [0,1]}$$

- Minimum value 0
  - Due to Nyquist Criterion, $\boxed{T_s \geq \frac{1}{2W}}$ $\Rightarrow$ $\boxed{\alpha = 2WT_s - 1 \geq 0}$
- Maximum value 1
  - Why?



$$\boxed{\frac{1}{T_s} - W = 0} \Rightarrow \boxed{\alpha = 2WT_s - 1 \leq 1}$$

Raised Cosine Filter: Properties (1)

- Some simple properties
  - between

$$\boxed{\alpha = 2WT_s - 1}$$

$$\Rightarrow \boxed{W = \frac{\alpha+1}{2T_s} = \frac{\alpha+1}{2}R_s} \Rightarrow \boxed{\frac{R_s}{2} \le W \le R_s}$$

$$\Rightarrow \boxed{R_s = \frac{1}{T_s} = \frac{2}{\alpha+1}W} \Rightarrow \boxed{W \le R_s \le 2W}$$

Simple ones, but can easily get confused, stick to your intuition!

Raised Cosine Filter: Properties (2)

- Bandwidth efficiency
  - bit rate per bandwidth（bps/Hz）
  - In general, according to Nyquist Criterion

$$\boxed{\eta_b = \frac{R_b}{W} = \frac{R_s \log_2 |\mathcal{S}|}{W} \le 2\log_2 |\mathcal{S}|}$$

- The bandwidth efficiency of Raised Cosine Filter acing as the baseband pulse waveform

$$\boxed{\eta_b = \frac{R_s \log_2 |\mathcal{S}|}{W} = \frac{2\log_2 |\mathcal{S}|}{\alpha + 1}}$$

Raised Cosine Filter: Exercise (1)

- A baseband transmission system for a PCM source
  - If the bandwidth limitation is 40kHz, and use the symbol set with M=2, what are feasible values of the rolloff factor?
  - If M=4, what is the largest bandwidth required?
- Solution: First, data rate of a PCM source is 64kbps
  - M=2, then the symbol rate is

$$\boxed{R_s = \frac{1}{\log_2 2} R_b = 64k\text{Hz}}$$

Notice: Be complete

  - We also have the inequalities for the rolloff factor

$$\boxed{\frac{\alpha + 1}{2} 64 \le 40} \Rightarrow \boxed{0 \le \alpha \le \frac{80}{64} - 1 = 0.25}$$

Raised Cosine Filter: Exercise (2)

- M=4, then the symbol rate is

$$\boxed{R_s = \frac{1}{\log_2 4} R_b = 32kbps}$$

- We also have the following inequality for the rolloff factor

$$\boxed{W \le R_s = 32kHz}$$

Thank you !

(Q&A)

Reference (Bit to Symbol Mapping):
[1] J. Proakis, *Digital Communications*, 5th edition, Section 3.2
Reference (Baseband Waveform):
[1] J. Proakis, *Digital Communications*, 5th edition, Section 9.2.1